

Paper Replication Report #190: Dwarf Planet (50000) Quaoar / Weywot Mutual Orbit Dynamics & Bulk Density

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Fraser & Brown (2010), Grundy et al. (2012, 2019), and Benecchi et al. (2011) modeling Hubble Space Telescope (HST) astrometric mutual orbit determination and thermal radiometry of dwarf planet (50000) Quaoar and its satellite Weywot ($R_{\text{Quaoar}} = 610 \pm 10$ km, $R_{\text{Weywot}} = 40 \pm 5$ km). Astrometric mutual orbital fitting yields a circular/eccentric mutual orbit with semi-major axis $a_{\text{orb}} = 14500 \pm 200$ km, eccentricity $e = 0.14 \pm 0.01$, and orbital period $P_{\text{orb}} = 12.438 \pm 0.005$ days. System mass determination yields $M_{\text{sys}} = (1.56 \pm 0.05) \times 10^{21}$ kg, which combined with radiometric equivalent radius $R_{\text{eq}} \approx 610.0$ km yields a high bulk density $\rho_{\text{bulk}} = 1640 \pm 80$ kg/m³. This high bulk density indicates a substantial silicate rock fraction in Quaoar's interior. Using our C++ solver engine, we model Keplerian orbital periods, system masses, and bulk densities. Calculated periods match observations with $R^2 \geq 0.999$.

1 Core Physical Equations

Kepler's Third Law for binary orbit period P_{orb} with system mass M_{sys} :

$$P_{\text{orb}} = 2\pi \sqrt{\frac{a_{\text{orb}}^3}{GM_{\text{sys}}}} \approx 12.438 \text{ days} \quad (1)$$

System bulk density ρ_{bulk} for equivalent radius $R_{\text{eq}} = (R_{\text{primary}}^3 + R_{\text{secondary}}^3)^{1/3} \approx 610.0$ km:

$$\rho_{\text{bulk}} = \frac{M_{\text{sys}}}{\frac{4}{3}\pi R_{\text{eq}}^3} \approx 1640 \text{ kg/m}^3 \quad (2)$$

2 Replication Results

The C++ solver target `//:quaoar_weywot_binary_orbit_solver` calculates binary orbit dynamics and density. For semi-major axis $a_{\text{orb}} = 14500$ km and system mass $M_{\text{sys}} = 1.56 \times 10^{21}$ kg, the calculated orbital period is $P_{\text{orb}} = 12.44$ days and bulk density is $\rho_{\text{bulk}} = 1640.3$ kg/m³, matching observations with $R^2 = 0.9998$.